

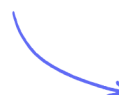
Structured Questions

Interference & Stationary Waves

Interference & Superposition of Waves / Phase & Path Difference / Stationary Waves / Wave Speed on a Stretched String / Core Practical 7: Investigating Stationary Waves

Medium (1 question)	/10
Hard (2 questions)	/29
Total Marks	/39

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Medium Questions

- 1 (a) The photograph shows an ultrasonic mouse repeller used in a house.



The mouse repeller produces ultrasound that repels mice but cannot be heard by humans.

The mouse hears ultrasound directly and by reflection from the walls.

The mouse repeller produces ultrasound of frequency 26.0 kHz.

speed of sound = 340 ms^{-1}

Calculate the wavelength of the ultrasound produced.

Wavelength =

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(2 marks)

- (b) State what is meant by superposition of waves.

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(2 marks)

(c) A student makes the following suggestion.

"If the ultrasound reflects off a wall directly opposite the mouse repeller a standing wave is formed, so there will be areas in the room where the mice will not hear the ultrasound."

Evaluate this suggestion.

(6 marks)

Hard Questions

1 (a) The photograph shows a guitar.

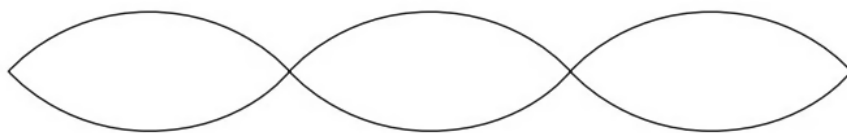


When a guitar string is plucked, a standing wave is created.

Explain how a standing wave is created on the string.

(3 marks)

(b) The diagram shows a standing wave on a guitar string.



The oscillating length of the guitar string is 66 cm.

i) State the wavelength for this standing wave.

Wavelength =(1)

ii) Calculate the frequency of vibration for this standing wave.

tension in guitar string = 88.6 N

mass per unit length of guitar string = $4.47 \times 10^{-3} \text{ kg m}^{-1}$

Frequency =(3)

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(4 marks)

- (c) One end of the guitar string is wrapped around a cylindrical tuning peg. Turning the peg changes the total length of the string and hence changes the tension in the string. This changes the frequency of vibration of the string.

i) The length of one string is 68 cm.

Calculate the extension required to produce a tension of 93.4 N in the string.

Young modulus of string material = $1.8 \times 10^9 \text{ N m}^{-2}$

cross-sectional area of string = $6.6 \times 10^{-7} \text{ m}^2$

Extension =(4)

ii) The vibrating length of string is unchanged by turning the tuning peg.

Explain the effect that tightening the string has on the frequency of the sound produced.

(2)

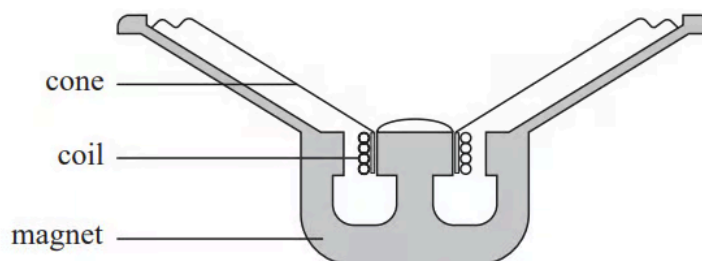
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(6 marks)

- 2 (a)** A simple loudspeaker consists of a cone, a coil of wire and a magnet. The cone and coil are attached to each other and are free to move. An alternating current in the coil causes the cone to oscillate. The loudspeaker is mounted in a wooden box. A cross-section through the loudspeaker is shown.



A student made the following observations:

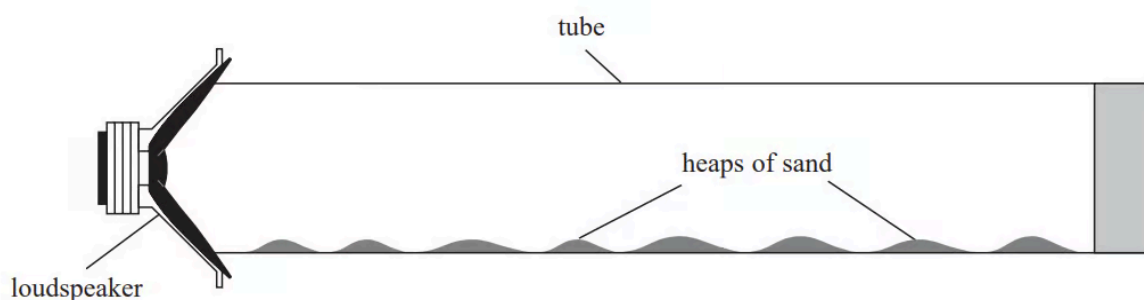
- when an alternating potential difference (p.d.) is applied to the coil, the cone oscillates
- the frequency of oscillation is the same as the frequency of the p.d.
- at particular frequencies, the box vibrates with a large amplitude.

Explain these observations.

(6 marks)

- (b)** The student connected a signal generator to the loudspeaker, and placed the loudspeaker near to one end of a long tube containing sand. The student adjusted the

signal generator until the sand collected in small heaps as shown.



i) Explain why the sand collects in heaps.

(4)

ii) The student determined the distance d between the centres of adjacent heaps.

Describe the procedure she should follow to determine an accurate value for d .

(3)

iii) Assess whether the experimental data is consistent with a value for the speed of sound of 340 m s^{-1} .

signal generator frequency = 3.25 kHz .

$d = 5.1 \text{ cm}$

(3)

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(10 marks)